

PROJECT REPORT

EMBEDDED SYSTEMS & IoT DEVICE DESIGN

Project Title: Smart Street Light
Automation System (LDR Upgrade)

Task Number: Embedded System Task-3

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Role: Junior Embedded Engineer (Intern)

SYSTEM SPECIFICATIONS & STACK

1. SYSTEM SPECIFICATIONS & COMPONENT MAPPING

The execution of this smart automation task utilizes a specific set of hardware behaviors modeled precisely within the virtual engine environment. Below are the design configurations:

1.1 Core Hardware Stack:

Microcontroller Board: ESP32 DevKit v4 (Selected for integrated Wi-Fi stack compatibility for future Cloud tasks)

Ambient Light Sensor: 4-Pin LDR (Photoresistor) Module with integrated voltage divider circuits

Output Actuator: High-Efficiency Red LED (Simulating a local automated street light node)

Current Limiter: 220 Ω Resistor (Connected inline with the LED cathode for standard terminal protection)

1.2 Software & Environment Stack:

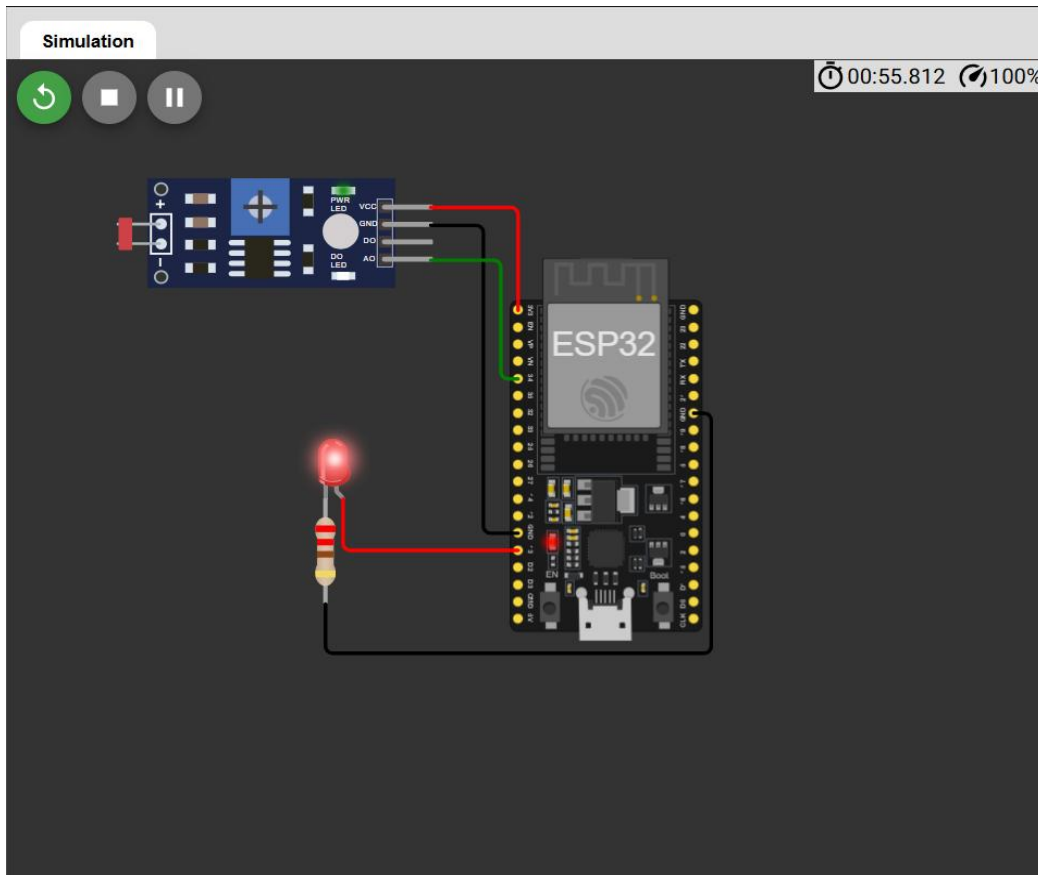
Development Environment: Wokwi Online Virtual Simulation Framework

Core Engine Language: Embedded C++ (Compiled on the native Arduino-ESP32 Core API layer)

Communication Port: Virtual UART via Serial Port (Configured at 115200 Baud Rate for real-time telemetry logging)

TECHNICAL IMPLEMENTATION & CODE

2. CIRCUIT DIAGRAM



3. SOURCE CODE

```
const int LDR_PIN = 34; // LDR Analog pin GPIO 34
const int LED_PIN = 13; // Smart Light Pin GPIO 13
const int THRESHOLD = 2000;

void setup() {
```

```
Serial.begin(115200);

pinMode(LDR_PIN, INPUT);

pinMode(LED_PIN, OUTPUT);

Serial.println("Smart Lighting System Started!");
}

void loop() {

    int ldrValue = analogRead(LDR_PIN);

    Serial.print("LDR (Light) Value: ");

    Serial.println(ldrValue);

    // Automation Logic (IF / ELSE)

    if (ldrValue > THRESHOLD) {
        digitalWrite(LED_PIN, HIGH); // TURN LED ON

        Serial.println("Status: [ANDHERA] -> Smart Light ON");
    }

    else {

        digitalWrite(LED_PIN, LOW); // TURN LED OFF

        Serial.println("Status: [UJALA] -> Smart Light OFF");
    }

    Serial.println();

    delay(1000);
}
```

RESULTS & WORKING EXPLANATION

3. WORKING EXPLANATION

This project successfully demonstrates how an embedded system makes independent automation decisions based on dynamic environmental changes.

The system continuously reads analog voltage variations from the surrounding environment using a **4-pin LDR Sensor Module**. This real-time data stream is processed by the central core of the **ESP32 microcontroller**.

An **IF / ELSE automation logic loop** is executed programmatically:

- **When ambient light drops (Darkness simulated via slider):** The analog values read by the sensor cross the pre-defined safety threshold of 2000. The code satisfies the IF condition statement, dynamically executing a HIGH command on **GPIO 13** to immediately switch the **LED ON**.
- **When ambient light is sufficient (Daylight simulated via slider):** The analog inputs stabilize well below the threshold limit, shifting execution to the default ELSE block routine. This instantly pulls **GPIO 13** to LOW, turning the **LED OFF** to conserve power.

4. SERIAL MONITOR MONITORING & OUTPUT



```
LDR (Light) Value: 1001
Status: [UJALA] -> Smart Light OFF

LDR (Light) Value: 2713
Status: [ANDHERA] -> Smart Light ON
```

The screenshot shows a serial monitor interface with a light gray background. It displays two sets of sensor data. The first set shows a value of 1001 and the status 'UJALA' leading to 'Smart Light OFF'. The second set shows a value of 2713 and the status 'ANDHERA' leading to 'Smart Light ON'. At the bottom right, there is a toolbar with icons for a graph, pause, clear, and a dropdown menu.